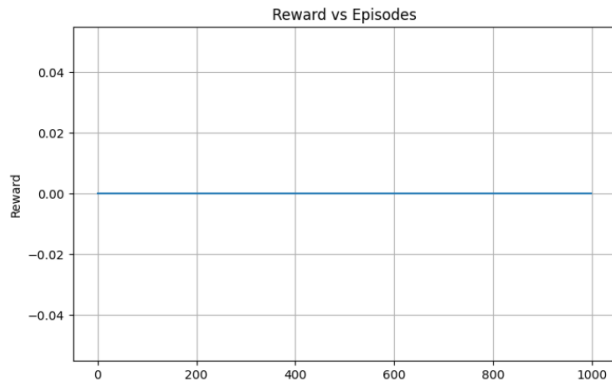


Optimal path:
[0, 0]



```
[7] ✓ Os
import numpy as np
from gym import Env, spaces
from gym.spaces import Discrete, Dict, Tuple

class WarehouseRobotEnv(Env):
    """Warehouse Robot Environment"""

    def __init__(self):
        """Initialize the environment"""
        self.action_space = Discrete(5)
        self.observation_space = Dict({
            'state': Discrete(5),
            'action': Discrete(5)
        })
        self.reset()

    def reset(self):
        """Reset the environment"""
        self.state = 0
        self.action = None
        self.reward = 0
        return self.observation_space.sample()

    def step(self, action):
        """Step the environment"""
        self.action = action
        self.state = (self.state + action) % 5
        self.reward = 0
        if self.state == 4:
            self.reward = 30
        return self.observation_space.sample(), self.reward, False, {}

    def render(self, mode='human'):
        """Render the environment"""
        print(f"State: {self.state}, Action: {self.action}, Reward = {self.reward}")

# Create the environment
env = WarehouseRobotEnv()

# Run the environment
for i in range(10):
    observation, reward, done, info = env.step(env.action_space.sample())
    print(f"Step {i}: Observation = {observation}, Reward = {reward}, Done = {done}")
    if done:
        break
```

[8]

```

import numpy as np

# -----
# States
# -----
states = ["Warehouse", "Road A", "Road B", "Customer"]

V = np.zeros(len(states))

# Discount Factor
gamma = 0.9

# Rewards
rewards = {
    0: -1, # Warehouse
    1: -2, # Road A
    2: -3, # Road B
    3: 100 # Customer (Goal)
}

# Transition Model
# Current State -> Next State
transitions = {
    0: [1, 2], # Warehouse -> Road A or Road B
    1: [3],    # Road A -> Customer
    2: [3],    # Road B -> Customer
    3: [],     # Goal
}

```

```

print(states[current], end="")
while current != 3:
    next_states = transitions[current]
    best = max(next_states, key=lambda x: V[x])
    print(" -->", states[best], end="")
    current = best
print("\n\nDelivery Completed Successfully!")

... Optimal State Values:
Warehouse : 78.20
Road A : 88.00
Road B : 87.00
Customer : 100.00

Optimal Path:
Warehouse --> Road A --> Customer
Delivery Completed Successfully!

```

```

... Estimated Click Rates (Q-values):
Ad 1: 0.082
Ad 2: 0.182
Ad 3: 0.143
Ad 4: 0.382
Ad 5: 0.343

```

Best Advertisement: 4
Total Clicks: 358

